space. He notices that Gracie's clock is running slow in comparison to his (with the rate of slowdown depending on how fast they pass one another). Were he a bit more astute, he would also note that in addition to the passage of time on her clock, everything about Gracie—the way she waves as she passes, the speed with which she blinks her eyes, and so on—is occurring in slow motion. From Gracie's perspective, exactly the same observations apply to George.

Although this seems paradoxical, let's try to pinpoint a precise experiment that would reveal a logical absurdity. The simplest possibility is to arrange things so that when George and Gracie pass one another they both set their clocks to read 12:00. As they travel apart, each claims that the other's clock is running slower. To confront this disagreement head on, George and Gracie must rejoin each other and directly compare the time elapsed on their clocks. But how can they do this? Well, George has a jetpack that he can use, from his perspective, to catch up with Gracie. But if he does this, the symmetry of their two perspectives, which is the cause of the apparent paradox, is broken since George will have undergone accelerated, non-force-free motion. When they rejoin in this manner, less time will indeed have elapsed on George's clock as he can now definitively say that he was in motion, since he could feel it. No longer are George's and Gracie's perspectives on equal footing. By turning on the jet-pack, George relinquishes his claim to being at rest.

If George chases after Gracie in this manner, the time difference that their clocks will show depends on their relative velocity and the details of how George uses his jet-pack. As is by now familiar, if the speeds involved are small, the difference will be minuscule. But if substantial fractions of light speed are involved, the differences can be minutes, days, years, centuries, or more. As one concrete example, imagine that the relative speed of George and Gracie when they pass and are moving apart is 99.5 percent of light speed. Further, let's say that George waits 3 years, according to his clock, before firing up his jet-pack for a momentary blast that sends him closing in on Gracie at the same speed that they were previously moving apart, 99.5 percent of light speed. When he reaches Gracie, 6 years will have elapsed on his clock since it will take him 3 years to catch her. However, the mathematics of special relativity shows that 60 years will have elapsed